

DAX Functions Documentation - Supermarket Sales Analysis

Dashboard 1: Sales Performance & Operations Analytics

1. KPI Metrics

DAX - Code

// Total Revenue

Total Revenue =

```
CALCULATE(  
    SUM('supermarket_sales'[Total])  
)
```

// Total Products Sold

Total Products =

```
CALCULATE(  
    SUM('supermarket_sales'[Quantity])  
)
```

// Average Transaction Value

Average Transaction Value =

```
DIVIDE(  
    SUM('supermarket_sales'[Total]),  
    DISTINCTCOUNT('supermarket_sales'[Invoice ID])  
)
```

// Gross Profit Margin

Gross Profit Margin =

```
DIVIDE(  
    SUM('supermarket_sales'[gross income]),  
    SUM('supermarket_sales'[Total]),  
    0
```

) * 100

2. Sales Analysis Measures

DAX - Code

// Daily Revenue

Daily Revenue =

```
CALCULATE(  
    SUM('supermarket_sales'[Total]),  
    VALUES('supermarket_sales'[Date])  
)
```

// Daily Quantity

Daily Quantity =

```
CALCULATE(  
    SUM('supermarket_sales'[Quantity]),  
    VALUES('supermarket_sales'[Date])  
)
```

// Revenue Moving Average

Revenue Moving Average =

```
AVERAGEX(  
    DATESINPERIOD(  
        'supermarket_sales'[Date],  
        LASTDATE('supermarket_sales'[Date]),  
        -7,  
        DAY  
    ),  
    [Daily Revenue]  
)
```

3. Branch Analysis

DAX - Code

// Branch Revenue

Branch Revenue =

```
CALCULATE(  
    SUM('supermarket_sales'[Total]),  
    VALUES('supermarket_sales'[Branch])  
)
```

// Branch Product Revenue

Branch Product Revenue =

```
CALCULATE(  
    SUM('supermarket_sales'[Total]),  
    VALUES('supermarket_sales'[Branch]),  
    VALUES('supermarket_sales'[Product line])  
)
```

// Branch Performance Index

Branch Performance Index =

```
DIVIDE(  
    [Branch Revenue],  
    CALCULATE(  
        [Branch Revenue],  
        ALL('supermarket_sales'[Branch])  
    ),  
    0  
)
```

Dashboard 2: Customer Insights & Product Analytics

1. Customer Rating Measures

DAX - Code

// Average Customer Rating

Average Customer Rating =

```
CALCULATE(  
    AVERAGE('supermarket_sales'[Rating])  
)
```

// Product Line Rating

Product Line Rating =

```
CALCULATE(  
    AVERAGE('supermarket_sales'[Rating]),  
    VALUES('supermarket_sales'[Product line])  
)
```

// Rating Trend

Rating Trend =

VAR CurrentRating = [Average Customer Rating]

VAR PrevRating =

```
CALCULATE(  
    [Average Customer Rating],  
    DATEADD('supermarket_sales'[Date], -1, MONTH)  
)
```

RETURN

```
IF(  
    NOT ISBLANK(PrevRating),  
    DIVIDE(CurrentRating - PrevRating, PrevRating, 0)  
)
```

2. Customer Segmentation Measures

DAX - Code

// Member Ratio

Member Ratio =

```
DIVIDE(  
    CALCULATE(  
        COUNTROWS('supermarket_sales'),  
        'supermarket_sales'[Customer type] = "Member"  
    ),  
    COUNTROWS('supermarket_sales'),  
    0  
)
```

// Customer Value

Customer Lifetime Value =

```
VAR AvgPurchase = DIVIDE(  
    SUM('supermarket_sales'[Total]),  
    DISTINCTCOUNT('supermarket_sales'[Invoice ID])  
)  
VAR PurchaseFreq = DIVIDE(  
    COUNTROWS('supermarket_sales'),  
    DISTINCTCOUNT('supermarket_sales'[Customer type])  
)  
RETURN  
AvgPurchase * PurchaseFreq
```

3. Product Performance Measures

DAX - Code

// Product Performance Index

Product Performance Index =

```
VAR Rating = [Product Line Rating]
```

```

VAR Revenue = [Product Line Revenue]
VAR MaxRevenue =
    CALCULATE(
        [Product Line Revenue],
        ALL('supermarket_sales'[Product line])
    )
VAR Margin = [Product Line Margin]
RETURN
(Rating/10 * 0.3) +
(DIVIDE(Revenue, MaxRevenue) * 0.4) +
(Margin * 0.3)

```

```

// Product Line Revenue
Product Line Revenue =
CALCULATE(
    SUM('supermarket_sales'[Total]),
    VALUES('supermarket_sales'[Product line])
)

```

```

// Product Line Margin
Product Line Margin =
DIVIDE(
    SUM('supermarket_sales'[gross income]),
    [Product Line Revenue],
    0
)

```

4. Time Analysis Measures

DAX - Code

```

// Hour Sales

```

```
Hour Sales =  
CALCULATE(  
    SUM('supermarket_sales'[Total]),  
    VALUES('supermarket_sales'[Hour])  
)  
  
// Day Type  
Day Type =  
IF(  
    WEEKDAY('supermarket_sales'[Date], 2) > 5,  
    "Weekend",  
    "Weekday"  
)
```

Common DAX Functions Used:

1. Aggregation Functions:

- AVERAGE
- SUM
- COUNT
- COUNTROWS
- DISTINCTCOUNT

2. Filter Functions:

- CALCULATE
- VALUES
- FILTER
- ALL

3. Logical Functions:

- IF
- SWITCH
- NOT
- ISBLANK

4. Mathematical Functions:

- DIVIDE
- MAX
- MIN

5. Time Intelligence:

- DATEADD
- DATESINPERIOD
- WEEKDAY